



Chapter 5 Gaussian Processes

- what do we have:

- inputs $x = [-1, 0.2]$
- basis/features $\phi(x) = [1, x]^T$
- design matrix $\Phi = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$
- prior on weights $w \sim \mathcal{N}(0, \alpha^{-1} I), w/\alpha = 1$
- noise model $\epsilon_n \sim \mathcal{N}(0, \sigma^2), w/\sigma = 2$
- Latent fun. values $f = \Phi w \rightarrow f_1 = w_1 - w_2, f_2 = w_2, f_3 = w_1 + w_2$

Weight-space perspective

a) Sample weights

- draw one realization from the prior, i.e. $w^{(1)} = \begin{bmatrix} 0.5 \\ -0.3 \end{bmatrix}$

b) compute the fun.

$$f^{(1)} = \Phi w^{(1)} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ -0.3 \end{bmatrix} = \begin{bmatrix} 1 \cdot 0.5 - (-0.3) \\ 0.5 \\ 1 \cdot 0.5 + (-0.3) \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.5 \\ 0.2 \end{bmatrix}$$

Function Space perspective

a) Derive the prior on f by integrating out w

- we have $f = \Phi w$

- $w \sim \mathcal{N}(0, I)$

$$w \perp f \quad p(f) = \int p(f|w) p(w) dw$$

- so $A = \Phi, \mu = 0, \Sigma = I, p(f|w) = \delta(f - \Phi w)$

$$\Rightarrow p(f) = \int \delta(f - \Phi w) \mathcal{N}(w|0, I) dw$$

- using the identity:

$$= \int \delta(f - Aw) \mathcal{N}(w|0, I) dw = \mathcal{N}(f|0, AA^T)$$

$$\therefore f \sim \mathcal{N}(0, \Phi I \Phi^T) = \mathcal{N}(0, \Phi \Phi^T) = \mathcal{N}(0, K), \text{ where } K = \Phi \Phi^T$$

$$K \Rightarrow \Phi \Phi^T = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

o sample directly from this prior

$$\text{- draw } f^{(1)} \sim \mathcal{N}(0, K) \Rightarrow f^{(1)} = \begin{bmatrix} 1.04 \\ 0.15 \\ -0.23 \end{bmatrix}$$

if, $w \sim \mathcal{N}(\mu, \Sigma)$ and $f = Aw + b$
then, $f \sim \mathcal{N}(A\mu + b, A\Sigma A^T)$

we use

$$p(z) = \mathcal{N}(z | \mu_z, \Sigma_z)$$

$$p(y|z) = \mathcal{N}(y | Wz + b, \Sigma_y)$$

then:

$$p(y) = \int p(y|z) p(z) dz = \mathcal{N}(y | W\mu_z + b, \Sigma_y + W\Sigma_z W^T)$$

Compute

$$p(y) = \int p(y|f) p(f) df$$

$$= \mathcal{N}(y | f, \beta^T I) \mathcal{N}(f | 0, K) df$$

$$= \mathcal{N}(y | ?, ?)$$

$$\Rightarrow \mathcal{N}(y | I \cdot 0 + 0, \beta^T I + 1 \cdot K \cdot 1^T)$$

$$\mathcal{N}(y | I \cdot 0 + 0, \beta^T I + K)$$

Gaussian processes detailed example

- training inputs : $x = [-1, 0, 1]^T$; $N=3$
- targets : $y = [-0.8, 0.1, 0.9]^T$; noisy sine data
- test input : $x^* = 0.5$
- Kernel k : squared exponential $\sigma_f^2 = 1$; $\ell = 1$; $k(x, x') = \exp[-(x-x')^2/2]$
- noise variance : $\sigma_v^2 = \sigma_n^2 = 0.3^2 = 0.09$

→ All prior means are 0 (classical GP regression)

Building the kernel matrix K for training set

- training inputs $x = [-1, 0, 1]^T$
- kernel $k(x, x') = \exp[-\frac{(x-x')^2}{2\ell^2}]$
- goal → fill 3×3 matrix K , whose (i, j) -entry is $k(x_i, x_j)$

i	j	pair (x_i, x_j)	dist d_{ij}	squared d_{ij}^2	exponent $-d_{ij}^2/(2\ell^2)$	$k(x_i, x_j)$
1	1	$(-1, -1)$	0	0	0	1
1	2	$(-1, 0)$	-1	1	-0.5	0.6
1	3	$(-1, 1)$	-2	4	-2	0.1
2	1	$(0, -1)$	+1	1	-0.5	0.6
2	2	$(0, 0)$	0	0	0	1
2	3	$(0, 1)$	-1	1	-0.5	0.6
3	1	$(1, -1)$	+2	4	-2	0.1
3	2	$(1, 0)$	1	1	-0.5	0.6
3	3	$(1, 1)$	0	0	0	1

- the following computes one similarity score between training inputs $x_i \wedge x_j$.

- it represents the prior covariance of the noise-free function values $f(x_i), f(x_j)$

- intuition: think of the system as a little Gaussian bump, centered at each training point. Comparing two input x_i, x_j , we ask:

How much overlap is there between the bump at x_i and a bump at x_j ?
 • $x_i = x_j \Rightarrow$ their overlap is covariance is 1
 • $x_i \neq x_j \Rightarrow$ overlap falls off smoothly and symmetrically like a bell curve. At a distance of one length-scale ($|x_i - x_j| = \ell$), the overlap is $\exp(-\frac{1}{2}) = 0.6$
 • further from x_j , the bump tapers off, so the prior says, those function values should be essentially uncorrelated.

Produces a following Kernel matrix

$$K = \begin{bmatrix} 1.0 & 0.6 & 0.1 \\ 0.6 & 1.0 & 0.6 \\ 0.1 & 0.6 & 1.0 \end{bmatrix}$$

- so now we choose the prior over latent functions. We picked squared kernel and 0 mean on the training inputs and build the $N \times N$ K matrix. it gives us a joint covariance of the noise-free latent values.

- Generative model $f \sim \mathcal{N}(0, K)$, $y = f + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma_v^2 I)$

- as next step we add the observational noise, this converts the latent-function prior into a prior over the observed targets y .

- marginalize out the latent f , gives us

$$y \sim \mathcal{N}(0, K + \sigma_v^2 I_N)$$

- computes: adds the observation noise variance $\sigma_v^2 = \sigma_n^2$ to the diagonal of K

- represents: the covariance of the **observed** targets $y = f + \epsilon$. Off-diagonals remain the same (noise is i.i.d.) diagonals increase.

- intuition: diagonal entries now say total uncertainty in each y_i : intrinsic function variance + measurement noise

- K represents our belief about how the **TRUE** functions values f co-vary

- Diagonal entries k_{ii} are $\text{Var}[f(x_i)] \rightarrow$ how uncertain you are about $f(x_i)$ itself
- Off-diagonals k_{ij} are $\text{Cov}[f(x_i), f(x_j)] \rightarrow$ how much a knowledge of one tells about the other.

- the observations $y_i = f(x_i) + \epsilon_i$ pick up extra variance $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$ indep $\forall i$
 ↳ this means
 - o each y_i is more uncertain, than $f(x_i)$, by exactly σ_ϵ^2 .
 - o the noise is $\perp$ across pts, so it does not increase any covariance between different y_i & y_j .
- if $\text{Cov}[f] = K$, $\text{Cov}[\epsilon] = \sigma_\epsilon^2 I$
- then by properties of covariance

$$\text{Cov}[y] = \text{Cov}[f + \epsilon] = \text{Cov}[f] + \text{Cov}[\epsilon] = K + \sigma_\epsilon^2 I$$

○ The cross-covariance vector $K_j = h(x^*, x)$

- computes len N row vector of similarities between the test input x^* and every training input
- represents how much the (latent) value $f(x^*)$ is covariant with each $f(x_i)$
- intuition: the larger a component K_j the more training pt. y_j will influence prediction

$$\text{ex } K = [k(x^*, x_1), k(x^*, x_2), k(x^*, x_3)] = [0.3, 0.8, 0.8]$$

$$x^* = 0.5 \Rightarrow h(0.5, 0) = \exp[-0.5^2/2] = 0.7$$

○ Prior Variance at x^* + noise

$$c = h(x^*, x^*) + \beta^{-1} = 1 + 0.09 = 1.09$$

- computes: prior fun. variance (which is 1 for SE kernel) + the some noise variance we used for training pts.
- represents: the variance of future noisy observation y^* Before we looked at any data
- intuition: if no data available, we would predict $y^* \sim \mathcal{N}(0, c) = \mathcal{N}(0, 1.09)$

○ Form the joint covariance of $\tilde{y} = \begin{bmatrix} f(x^*) \\ y \end{bmatrix}$

$$\tilde{C} = \begin{bmatrix} c & h^T \\ h & C \end{bmatrix} = \begin{bmatrix} 1.09 & 0.3 & 0.8 & 0.8 \\ 0.3 & 1.09 & 0.6 & 0.1 \\ 0.8 & 0.6 & 1.09 & 0.6 \\ 0.8 & 0.1 & 0.6 & 1.09 \end{bmatrix}$$

- o what $\tilde{y}$ is: y the N noisy targets we have already measured
 $y(x^*)$ the noisy target we have not measured yet at the new input x^* . It's still a random variable that could be observed in the future
- o computes: stacks the pieces you already have into (N+1)(N+1) matrix
- o represents: joint prior uncertainty of the whole vector of interest
- o upper-right block: h says "test as seen sampling"; lower right is the noise-inflated training block you believe until data are known
- o $C = K + \sigma_\epsilon^2 I$ (train-train cov)
 - covariance of the observed data y
 - how uncertain each measurement is in 2. how they relate to each other
- o $h = [h(x^*, x_1), \dots, h(x^*, x_N)]$ (test-train)
- cov. between the un-measured $y(x^*)$ and each measured y_j .
 - "Who are my useful-neighbors?" $\Rightarrow$ larger entry $\Rightarrow$ stronger influence
- o $C = h(x^*, x^*) + \sigma_\epsilon^2$ (test-variance)
 - prior variance of the future noisy observation at x^*
 - how uncertain we would be without any data

○ Conditioning formulas

□ Predictive mean $m_{y^*|y} = h C^{-1} y = h (K + \beta^{-1} I)^{-1} y$

- o computes:
 - $w = h C^{-1} \Rightarrow$ a weight row that sums to < 1
 - Dot product $w y = h C^{-1} y \Rightarrow$ single scalar
- o Represents posterior best guess of y^* after seeing data
- o Intuition:
 - w_j is large if:
 - K_j is large (point is near x^*), and
 - row/column j of C has small variance (low noise or many neighbors)
 - $\therefore m_{y^*|y}$ is literally a weighted average of training targets, w/ proximity-and-noise-aware weights

□ Predictive variance $\sigma_{y^*|y}^2 = c - h C^{-1} h^T = c - h (K + \beta^{-1} I)^{-1} h^T$

- o computes: subtracts a non-negative scalar from the prior variance c .
- o represents: Posterior uncertainty in the prediction (includes noise)

- Intuition:
 - The term $\mathbf{h}^T \mathbf{C}^{-1} \mathbf{h}$ is how much the data explain the prior variance.
 - If ϵ larger when the test pt. is strongly correlated w/ reliable training points, hence uncertainty shrinks the most near dense low-noise regions.

Why $\mathbf{C}^{-1}$ (the precision) shows up

- in a multivar. normal, information not covariance determines conditioning. $\mathbf{C}^{-1}$ down-weights pts. w/ high own variance or strong correlations that create redundancy.
- Computationally, we rarely form $\mathbf{C}^{-1}$ explicitly: we solve $\mathbf{C}\mathbf{w} = \mathbf{h}^T$ (Cholesky) to obtain $\mathbf{w} = \mathbf{h} \mathbf{C}^{-1}$ more stably.

Summary: think of $\mathbf{h}$ as a hand reaching from the test pt. to pull information from each training pt. The closer the pt. the stronger the pull.

- think of $\mathbf{C}^{-1}$ as a filter that says "I trust some of these pts. more than others, base of noise and redundancy.
- the mean is where that hand lands after being filtered.
- the variance is how much uncertainty is left after the pull.
- $p(y^* | y, x^*) = \mathcal{N}(y^* | \mu_{y^*|y}, \sigma_{y^*|y}^2)$

How the "joint-then-slice" trick work

- for any jointly Gaussian vector,

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}\right), \text{ the conditional distribution is available in closed form}$$

$$p(y_1 | y_2) = \mathcal{N}\left(\Sigma_{12} \Sigma_{22}^{-1} y_2, \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}\right)$$

So once we have the joint $\tilde{\mathbf{C}}$, we apply this universal formula w/

- $y_1 \equiv y(x^*)$
- $y_2 = y$
- block $(\Sigma_{11}, \Sigma_{12}, \Sigma_{21}) = (\mathbf{C}, \mathbf{h}, \mathbf{C})$

Reading

○ Marginal & Conditional

- led partition the random vector as

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}; \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}; \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}, y_1 \in \mathbb{R}^{d_1}, y_2 \in \mathbb{R}^{d_2}, d_1 + d_2 = D$$

- led the precision matrix be $\Lambda = \Sigma^{-1}$ and partition like $\Lambda = \Sigma^{-1} = \begin{bmatrix} \Lambda_{11} & \Lambda_{12} \\ \Lambda_{21} & \Lambda_{22} \end{bmatrix}$

- shapes:
 - $\Sigma_{11} \in \mathbb{R}^{d_1 \times d_1}$
 - $\Sigma_{22} \in \mathbb{R}^{d_2 \times d_2}$ (symmetric, PD)
 - $\Sigma_{12} = \Sigma_{21}^T \in \mathbb{R}^{d_1 \times d_2}$

EX: $d_1 = 2, d_2 = 3, D = d_1 + d_2 = 2 + 3 = 5$

$y_1 = (2, 1)$	$\Sigma_{11} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$
$y_2 = (3, 1)$	$\Sigma_{22} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$
$\mu_1 = (2, 1)$	$\Sigma_{12} = \begin{pmatrix} 2 & 3 \\ 2 & 3 \end{pmatrix}$
$\mu_2 = (3, 7)$	$\Sigma_{21} = \Sigma_{12}^T = \begin{pmatrix} 2 & 3 \\ 2 & 3 \end{pmatrix}$

- marginals are trivial (we just pick the corresponding block)

$$y_1 \sim \mathcal{N}(\mu_1, \Sigma_{11}), y_2 \sim \mathcal{N}(\mu_2, \Sigma_{22})$$

- conditionals are Gaussian $p(y_1 | y_2) = \mathcal{N}(y_1 | \mu_{1|2}, \Sigma_{1|2})$

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \quad \vee \quad \Sigma_{1|2} = \Lambda^{-1}_{11}$$

$$\mu_{1|2} = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (\mu_2 - \mu_2) \quad \vee \quad \mu_{1|2} = \mu_1 - \Lambda_{12}^{-1} \Lambda_{22} (\mu_2 - \mu_2) = \Sigma_{12} (\Lambda_{11} \mu_1 - \Lambda_{12} (\mu_2 - \mu_2))$$



Task 1.1 Show that if $z \sim \mathcal{N}(0, I)$ and $m \in \mathbb{R}^N$, then $f = Lz + m \sim \mathcal{N}(m, K)$.

$$E[f] = E[Lz + m] = E[Lz] + E[m] = L \cdot E[z] + m = L \cdot 0 + m = m$$

- using rule $E[\tilde{a}] = a$

$$E[ax] = aE[x]$$

$$E[ax + by] = aE[x] + bE[y]$$

$$V[f] = V[Lz + m] = L V[z] L^T = L L^T = K$$

- using rule $\text{Var}(aX + b) = a \text{Var}(X) a^T$

Task 1.2 Prove that adding a jitter increases the eigenvalues of the resulting matrix

- we know $K \in \mathbb{R}^{N \times N}$ (symmetric)
- (λ, v) is an e-val.-vec. pair of K , i.e. $Kv = \lambda v$
- I_n is the $N \times N$ id. mat. w.t. v , $I_n v = v$
- $\epsilon > 0$ is the jitter added to the diagonal

let $K + \epsilon I$. Apply the same to the eigenvector v .

$$(K + \epsilon I)v = Kv + \epsilon I_n v = \lambda v + \epsilon v = (\lambda + \epsilon)v$$

2.3 How does parameters b, ℓ affect the structure of the kernel?

$$K_{mm} = k_{ss}(x_n, x_m) = b^2 \exp\left(-\frac{(x_n - x_m)^2}{2\ell^2}\right) \quad m, m = 1, \dots, N$$

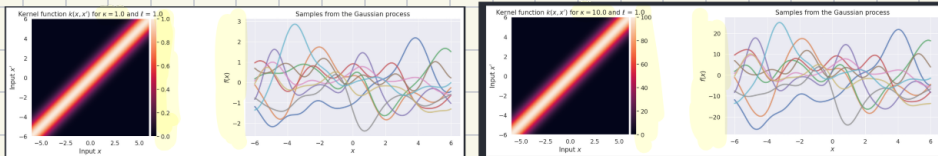
- exponential prefactor
 - controls how fast k decays with distance
 • small ℓ : rapid decay, thin band on negative diagonal
 • big ℓ : slow decay, thick band on negative diagonal

o effect of b , fix ℓ

$$k = b^2 e^{-d^2/2\ell^2} \Rightarrow \frac{\partial k}{\partial b} = 2b e^{-d^2/2\ell^2}$$

The derivative $\frac{\partial k}{\partial b} > 0$ everywhere except 0, so increasing b raises the whole kernel surface

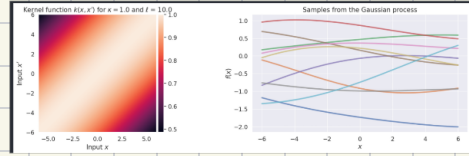
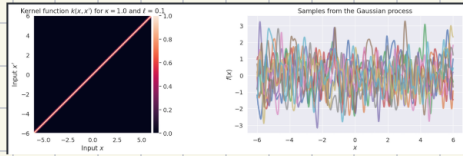
Figure below: the colour map is rescaled from $b_{\text{old}}^2 = 1.0$ to $b_{\text{new}}^2 = 10.0$ every pixel grows, but shape is unchanged.



Effect of ℓ , we fix h and vary ℓ .

$$k = h^2 \exp\left[-\frac{1}{2} \left(\frac{d}{h}\right)^2\right]$$

- we can rewrite $r = \frac{1}{h^2}$. As h increases, r decreases $\forall d > 0$, and the kernel values increase towards the values of the diagonal.



Task 2.4 How does h and ℓ affect prior samples

- how h and ℓ shape the prior samples

We draw $f \sim \mathcal{N}(0, K)$, generated by $f = Lz$ $K_{m,n} = h_{SE}(x_n, x_m) = h^2 \exp\left(-\frac{(x_n - x_m)^2}{2\ell^2}\right)$
 $(\hookrightarrow LL^T = K \quad z \sim \mathcal{N}(0, I)$

- h scales every variance and covariance by $\text{Var}[f(x)] = h^2$. The scale is vertical.
 - large h : taller peaks, and more pronounced dips.
 - small h : flatter, low-amplitude curves.
- ℓ controls correlation as a function of distance $d = |x - x'|$. Horizontal smoothness.
 - large ℓ : points far apart remain strongly correlated. Slow, smooth and broad.
 - small ℓ : correlation dies quickly. Rapid fluctuations and many wrinkles.

Task 2.5 How does samples from $M_{\frac{1}{2}} \sim M_{\frac{3}{2}}$ differ from SE?

- $M_{\frac{1}{2}}$: looks like cont. random walk, lots of kinks, roughest of the 3.
- $M_{\frac{3}{2}}$: wiggly w/ smooth curves,



2.6

Using the squared exponential covariance function and assume $\kappa = 1$, what is the covariance between the function values $f(\mathbf{x}_n)$ and $f(\mathbf{x}_m)$ for $d = 0, 1, 2, 3, 4, 5$?

$$k = k^2 \exp\left(-\frac{\|\mathbf{x}_n - \mathbf{x}_m\|}{2k^2}\right)$$

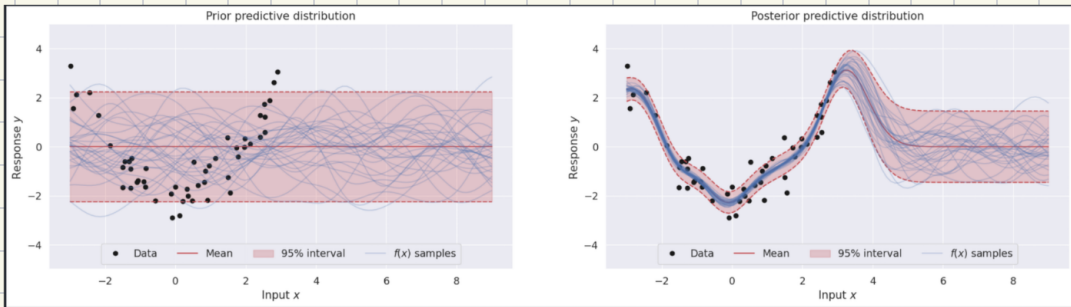
$$= 1 \cdot \exp\left(-\frac{(d \cdot 1)^2}{2 \cdot 1^2}\right)$$

$$= \exp\left(-\frac{d^2 \cdot 1^2}{2 \cdot 1^2}\right)$$

$$= \exp\left(-\frac{d^2}{2}\right)$$

At $d=1$ the correlation is 0.61. At $d=2$ it drops to 0.14.

As d increases, correlation decreases.



Coxe notation used in GP reg.

○ training data $X = \{x_1, \dots, x_n\} \in \mathbb{R}^{n \times D}$, $y \in \mathbb{R}^{n \times 1}$

○ $X^* = \{x_1^*, \dots, x_p^*\} \in \mathbb{R}^{p \times D}$

○ kernel hyper parameters $\theta = \{b, \ell\}$, ℓ is the std. or

□ Kernel matrices

- $K = K_{xx} = k(X, X) \in \mathbb{R}^{n \times n}$
- $b = K_{xx} = k(X^*, X) \in \mathbb{R}^{p \times n}$
- $L = K_{xx} = k(X^*, X^*) \in \mathbb{R}^{p \times p}$
- $C = K + \sigma^2 I_n$

○ posterior for latent fun. val

$$f_* | y = \mathcal{N}(\mu_*, \Sigma_*) \quad \begin{cases} \mu_* = b C^{-1} y \\ \Sigma_* = L - b C^{-1} b^T \end{cases}$$

○ posterior for latent fun. val

$$y_* | y = \mathcal{N}(\mu_*, \Sigma_* + \sigma^2 I_p)$$